

VZ-200/300

Memory Map.

VZ-200

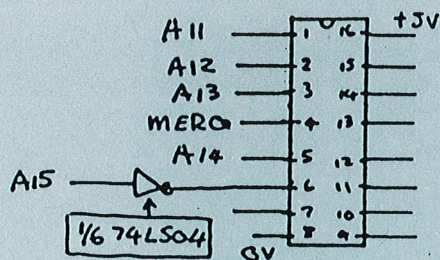
DECIMAL

HEX

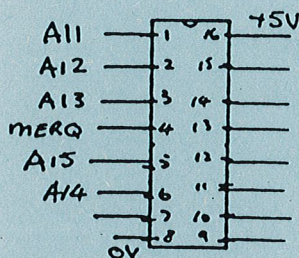
VZ-300

	65535	-1	^{F800} FFFF	
	61440	-4096	F000	
	57344	-8192	E000	
	53248	-12288	D000	16K RAM PACK (USER) (External)
	49152	-16384	C000	
	45056	-20480	B000	
16K RAM PACK (USER) (External)	40960	-24576	A000	
	36864	-28672	9000	16K USER RAM (INTERNAL)
	32768	-32768	8000	
	32767		7FFF	
6K USER RAM (Internal)			7800	
			77FF	
	28672	Video Ram 2k.	7000	
			6FFF	
		Keyboard, Cass, Spkr	6800	
			67FF	
	24576		6000	
			5FFF	
8K 'DISK' Rom	20480	4000 - 67FF	5000	8K 'DISK' Rom
	16384		4000	
		Rom 1		
	12288		3000	
		Rom 1		
16K 'BASIC' Rom	8192		2000	16K 'BASIC' Rom
		Rom 0		
	4096		1000	
		Rom 0		
	0		0000	

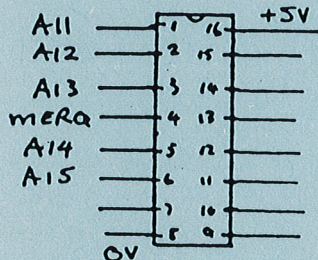
64K Decoding with 4x74LS138 in 2K Blocks



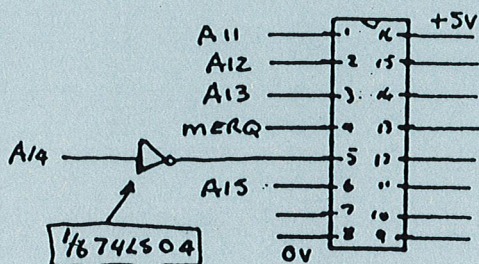
16K



32K



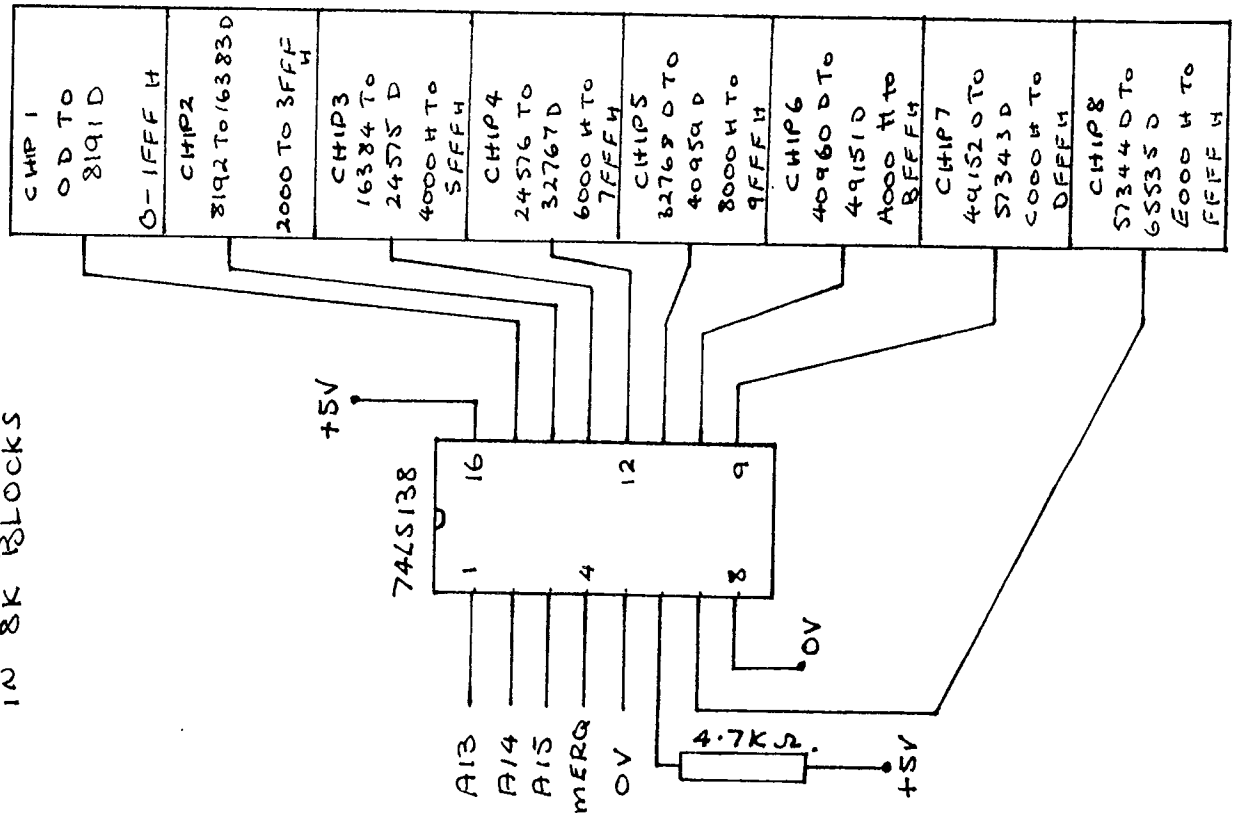
48K



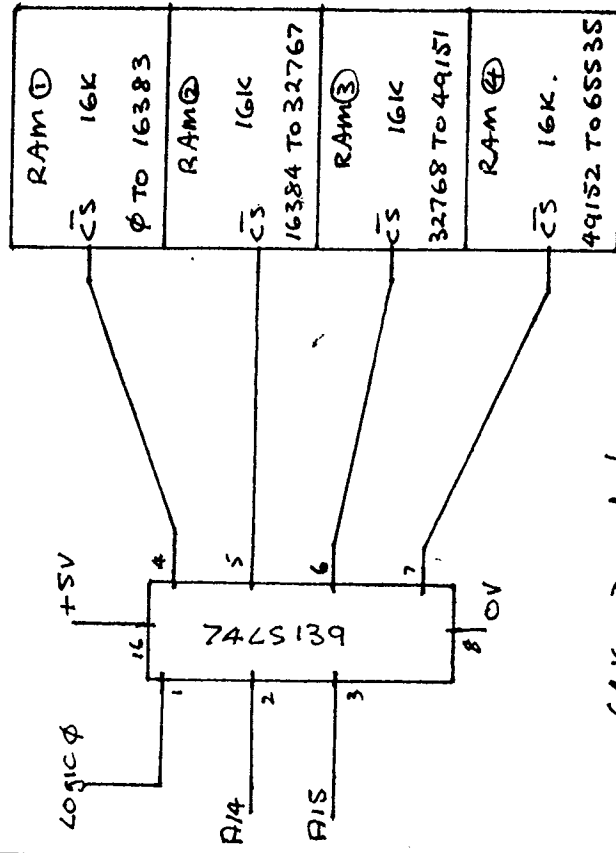
64K

PIN No	MEMORY Location
15	0000 - 07FF
14	0800 - 0FFF
13	1000 - 17FF
12	1800 - 1FFF
11	2000 - 27FF
10	2800 - 2FFF
9	3000 - 37FF
7	3800 - 3FFF
15	4000 - 47FF
14	4800 - 4FFF
13	5000 - 57FF
12	5800 - 5FFF
11	6000 - 67FF
10	6800 - 6FFF
9	7000 - 77FF
7	7800 - 7FFF
15	8000 - 87FF
14	8800 - 8FFF
13	9000 - 97FF
12	9800 - 9FFF
11	A000 - A7FF
10	A800 - AFFF
9	B000 - B7FF
7	B800 - BFFF
15	C000 - C7FF
14	C800 - CFFF
13	D000 - D7FF
12	D800 - DFFF
11	E000 - E7FF
10	E800 - EFFF
9	F000 - F7FF
7	F800 - FFFF

64K MEMORY IN 8K BLOCKS



64K MEMORY IN 16K BLOCKS



64K Decoded
IN 4 by 16K
Blocks.

using 1/2 74LS139.

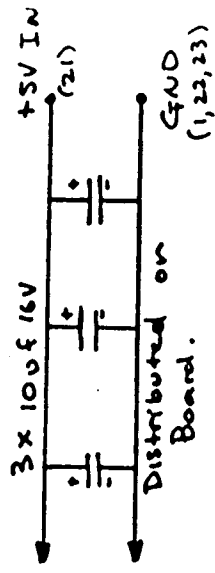
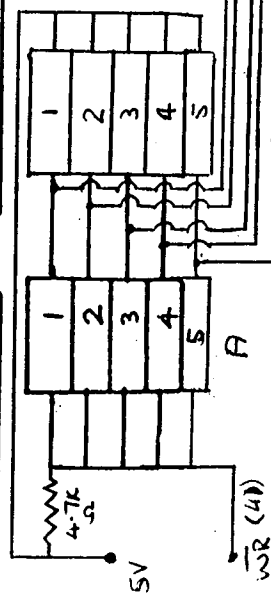
PROJECT

VZ-200 - RAM/EPROM BOARD (COPYRIGHT 1984 - DOZING SOFT/HARDWARE)

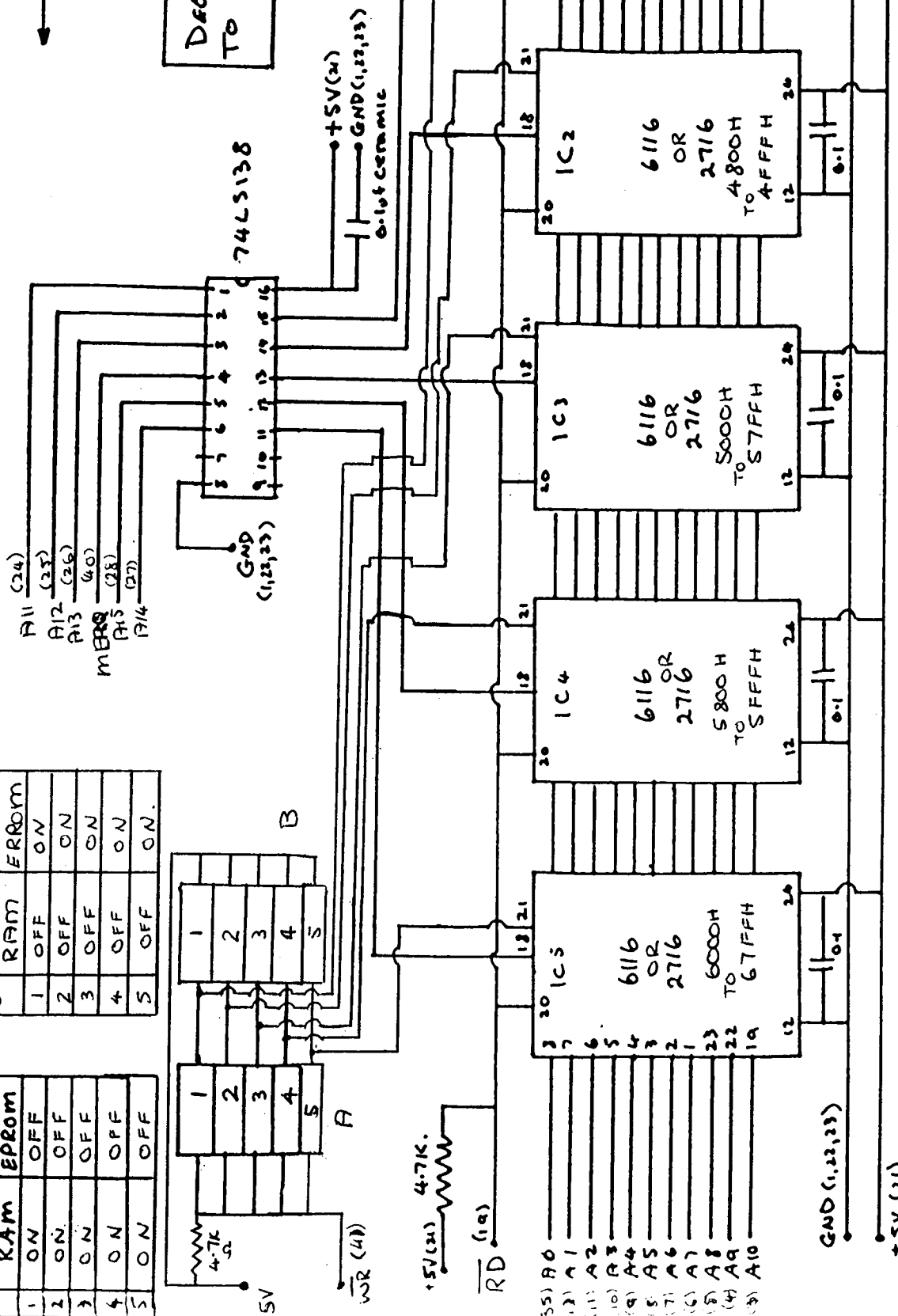
10K

SWAY DIP SWITCH	
A	RAM EPROM
1	ON OFF
2	ON OFF
3	ON OFF
4	ON OFF
5	ON OFF

5 WAY DIP SWITCH	
B	RAM EPROM
1	OFF ON
2	OFF ON
3	OFF ON
4	OFF ON
5	OFF ON

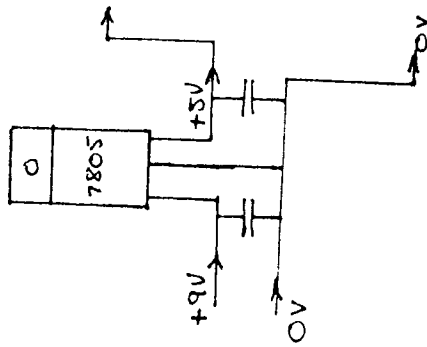


DECODING FROM 4000 HEX TO 67FF HEX IN 2K BLOCKS.



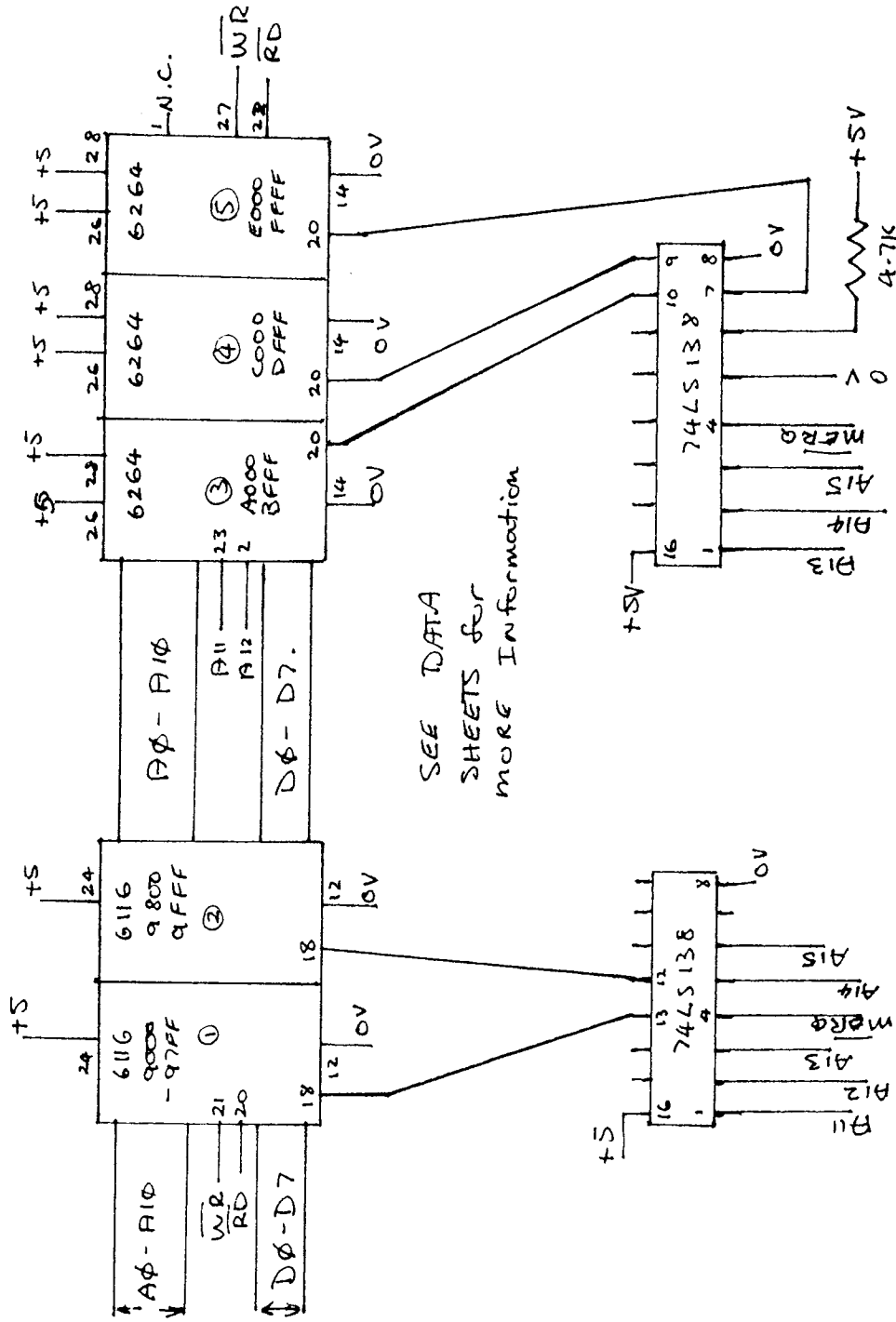
5x 0.1μfd ceramic at chip power pins

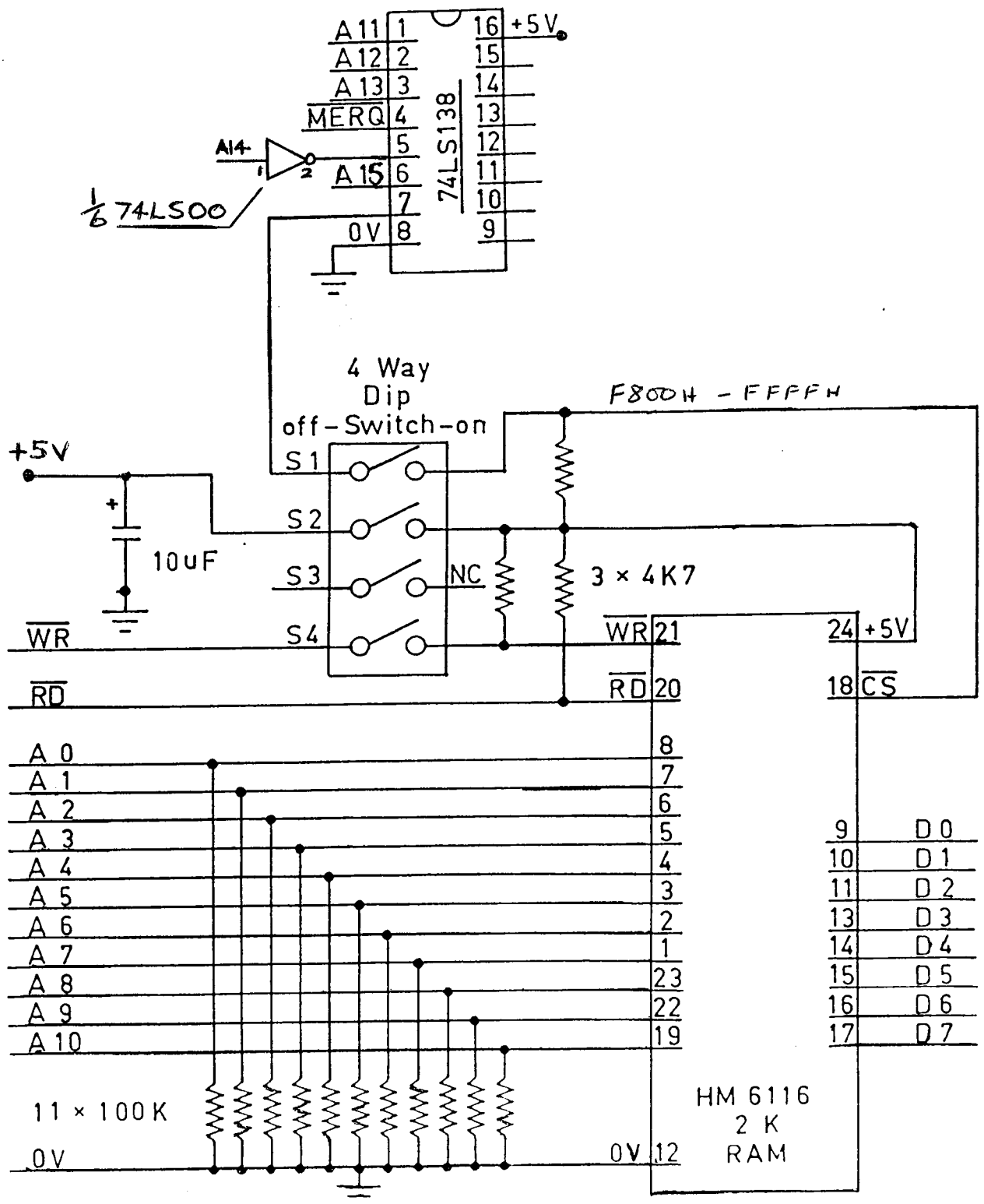
(*) - VZ-200 EXPANSION BUS CONNECTOR PIN NUMBERS.



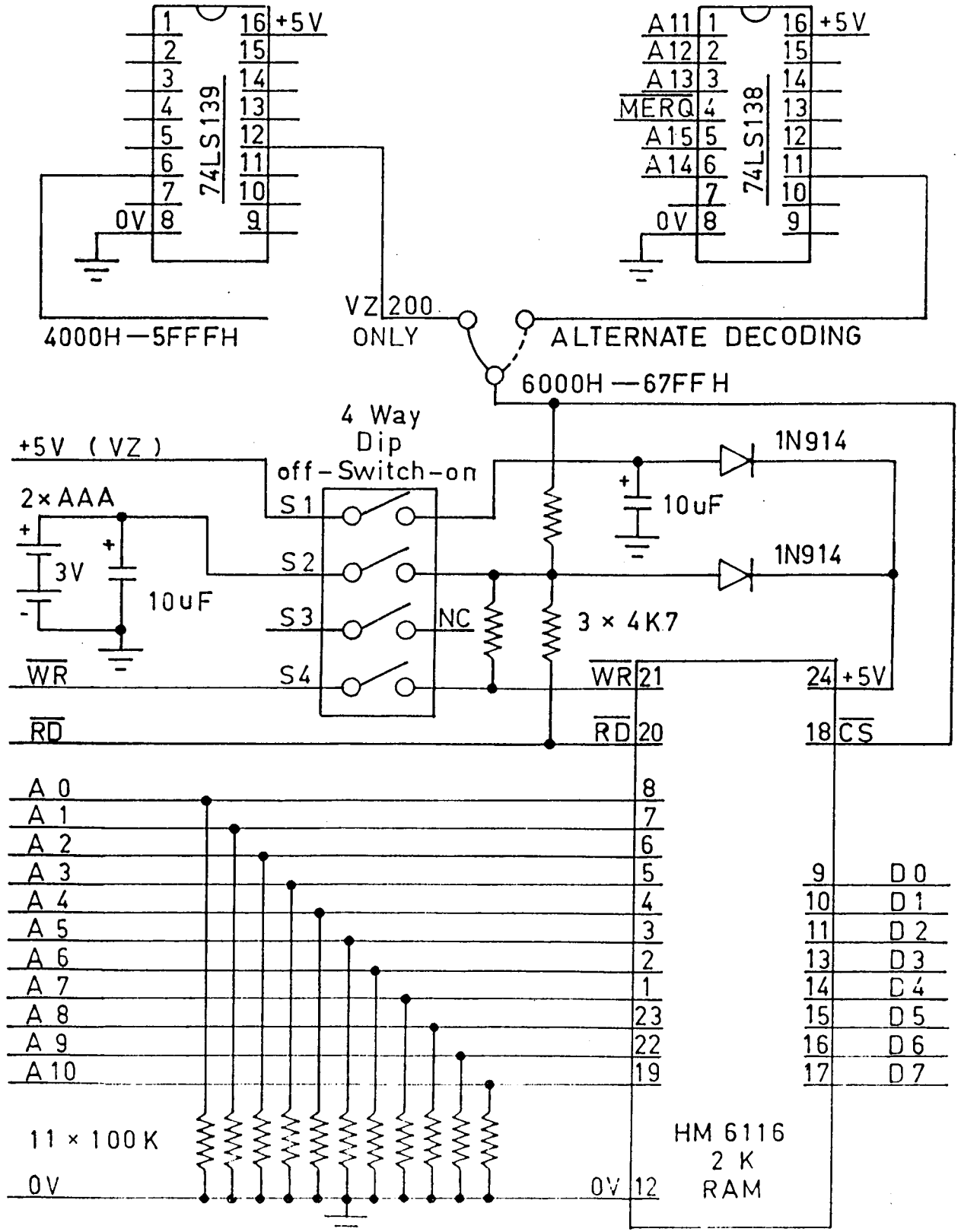
28K RAM
BOARD
DECODING
FROM
9000H TO
FFFFH

USING
2 X 6116
3 X 6264
2 X 74LS138
INSTALL a
0.01uF cap on
the +5 pin of each
chip.





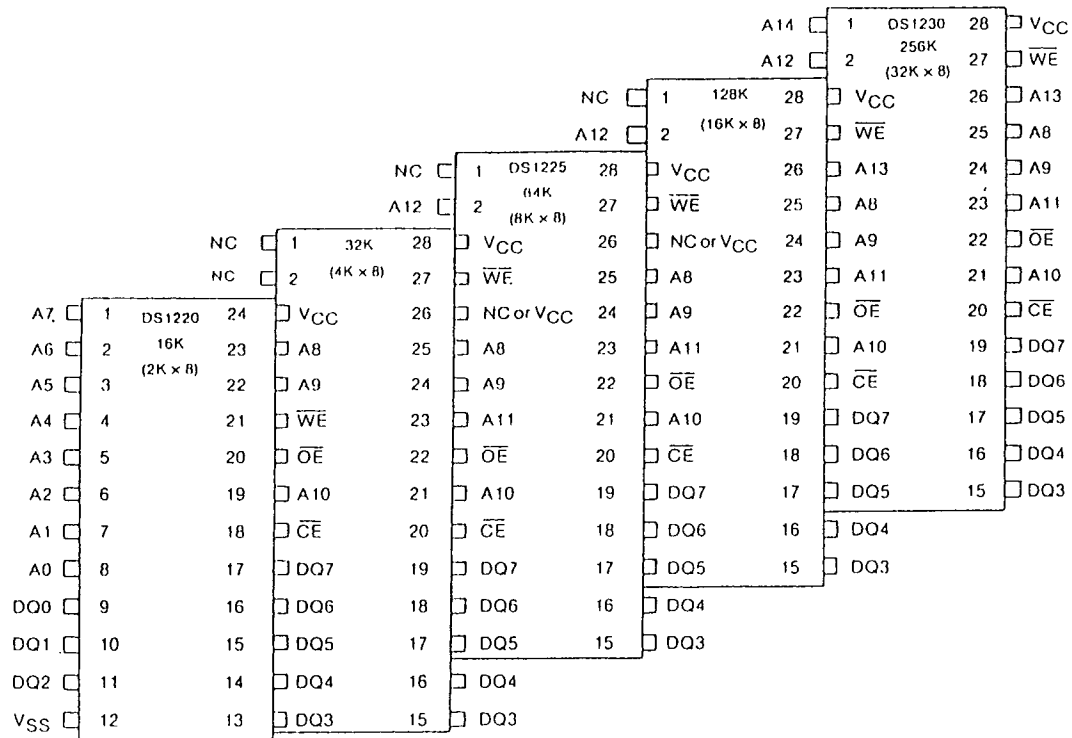
VZ 200 / 300 BATTERY BACK UP RAM



V2-200/300 Memory Map.

V2-200	DECIMAL	HEX	V2-300
	65535 -1	FFFF	
	61440 -4096	F000	
	57344 -8192	E000	
	53248 -12288	D000	16K RAM PACK (USER)
	49152 -16384	C000	
	45056 -20480	B000	
16K RAM PACK (USER)	40960 -24576	A000	
	36864 -28672	9000	16K USER RAM
	32768 -32768	8000	
6K USER RAM	32767	7FFF	
		7800	
	28672 Video Ram 2K.	7000	
		6FFF	
	Keyboard, Cass, Spkr	6800	
		67FF	
	24576	6000	
		5FFF	
8K DISK ROM	20480 4000 - 67FF	5000	8K DISK ROM
	16384	4000	
	Rom 1		
	12288	3000	
	Rom 1		
16K BASIC ROM	8192	2000	16K BASIC ROM
	Rom 0		
	4096	1000	
	Rom 0		
	0	0000	

Figure 6 **Byte-wide JEDEC Pinout**



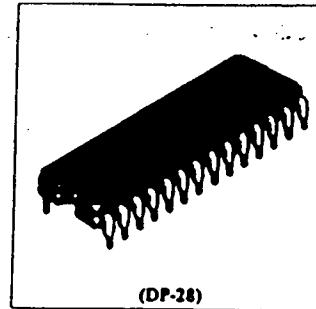
Provides for density upgrade to 32K x 8 RAM

HM6264P-10, HM6264P-12, HM6264P-15

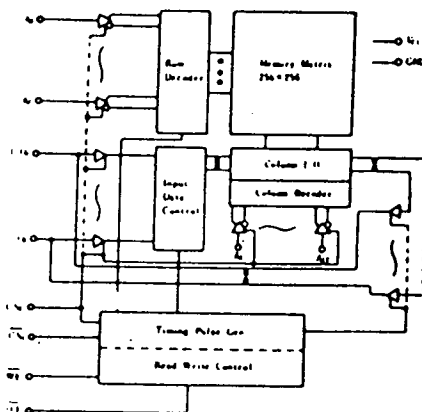
8192-word x 8-bit High Speed Static CMOS RAM

■ FEATURES

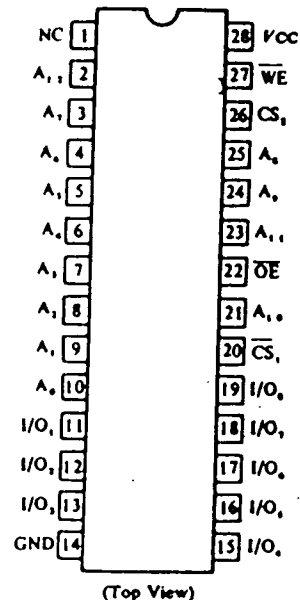
- Fast access Time 100ns/120ns/150ns (max.)
- Low Power Standby Standby: 0.1mW (typ.)
- Low Power Operation Operating: 200mW (typ.)
- Single +5V Supply
- Completely Static Memory. . . . No clock or Timing Strobe Required
- Equal Access and Cycle Time
- Common Data Input and Output, Three State Output
- Directly TTL Compatible: All Input and Output
- Standard 28pin Package Configuration
- Pin Out Compatible with 64K EPROM HN482764



■ BLOCK DIAGRAM



■ PIN ARRANGEMENT



■ ABSOLUTE MAXIMUM RATINGS

Item	Symbol	Rating	Unit
Terminal Voltage *	V_T	-0.5 ** to +7.0	V
Power Dissipation	P_T	1.0	W
Operating Temperature	T_{opr}	0 to +70	°C
Storage Temperature	T_{stg}	-55 to +125	°C
Storage Temperature (Under Bias)	T_{bias}	-10 to +85	°C

* With respect to GND. ** Pulse width 50ns: 3.0V

■ TRUTH TABLE

WE	CS ₁	CS ₂	OE	Mode	I/O Pin	I_{CC} Current	Note
X	H	X	X	Not Selected (Power Down)	High Z	I_{BB}, I_{BB1}	
X	X	L	X		High Z	I_{BB}, I_{BB2}	
H	L	H	H	Output Disabled	High Z	I_{CC}, I_{CC1}	
H	L	H	L	Read	Dout	I_{CC}, I_{CC1}	
L	L	H	H	Write	Din	I_{CC}, I_{CC1}	Write Cycle (1)
L	L	H	L		Din	I_{CC}, I_{CC1}	Write Cycle (2)

X: Don't care



PIN (23) 21 on 6116 is $\overline{WP}$
PIN (20) 18 on 6116 is chip select.
PIN 26 (6264) To +5V

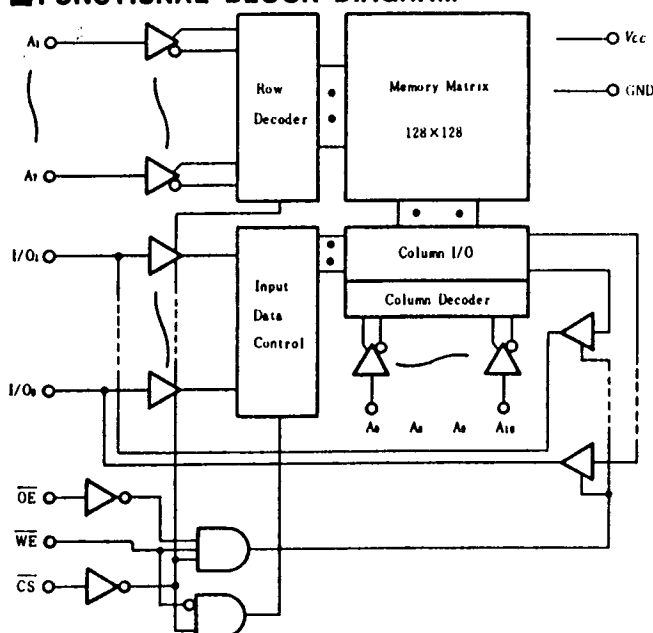
HM6116LP-2, HM6116LP-3, HM6116LP-4

2048-word $\times$ 8-bit High Speed Static CMOS RAM

FEATURES

- Single 5V Supply and High Density 24 Pin Package
- High Speed: Fast Access Time 120ns/150ns/200ns (max.)
- Low Power Standby and Standby: 10 μ W (typ.)
- Low Power Operation; Operation: 160mW (typ.)
- Completely Static RAM: No clock nor Timing Strobe Required
- Directly TTL Compatible: All Input and Output
- Pin Out Compatible with Standard 16K EPROM/MASK ROM
- Equal Access and Cycle Time
- Capability of Battery Back up Operation

FUNCTIONAL BLOCK DIAGRAM



ABSOLUTE MAXIMUM RATINGS

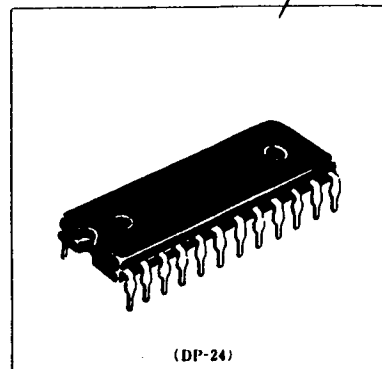
Item	Symbol	Rating	Unit
Voltage on Any Pin Relative to GND	V_T	-0.5* to +7.0	V
Operating Temperature	T_{op}	0 to +70	°C
Storage Temperature	T_{stg}	-55 to +125	°C
Temperature Under Bias	T_{mb}	-10 to +85	°C
Power Dissipation	P_T	1.0	W

* Pulse Width 50ns : -1.0V

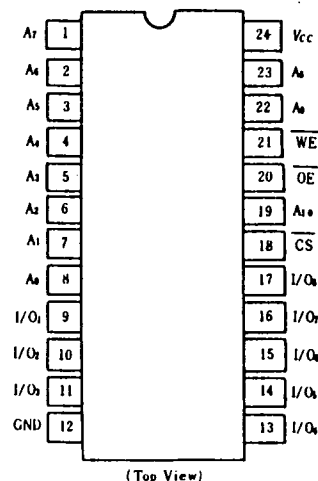
TRUTH TABLE

CS	OE	WE	Mode	V_{cc} Current	I/O Pin	Ref. Cycle
H	x	x	Not Selected	I_{sn}, I_{sb1}	High Z	
L	L	H	Read	I_{cc}	Dout	Read Cycle (1)~(3)
L	H	L	Write	I_{cc}	Din	Write Cycle (1)
L	L	L	Write	I_{cc}	Din	Write Cycle (2)

MASTER COPY



PIN ARRANGEMENT



ELECTRONICS & COMPUTING

Presented free with the October issue of Electronics and Computing Monthly.

The Centronics Parallel Interface

The majority of parallel printer interfaces adopt the Centronics standard for data exchange.

The interface is implemented on a 36 pin connector - the pin assignment is as shown in Table 1.

The data is transferred via lines 2-9 while signals at pins 1, 10 and 11 control the interchange of data.

To transfer the eight bit byte from source to destination (nominally from computer to printer) the data is first placed on lines 2-9. Line 1 (data strobe) is then taken low to signal that the data is ready. When the printer has received the data it will pull line 10, acknowledge, low.

These 10 signal lines (eight data and two control) are all that are needed for a basic parallel interface.

Of the other lines, the most significant is probably the busy, pin 11, signal. Often printers will not output characters until a full line has been received (a carriage return of line feed signal is detected). The printer then becomes 'busy' and some software will detect this condition and cause a computer to perform other tasks during the time that the printer is 'tied up'.

The function lines 12 and 13 are fairly self-explanatory.

For the most reliable data transfer, the connecting cable used should be of a twisted pair type - even so cable lengths should be limited to about two metres.

Communications Data Card

PIN SIGNAL	35 -	36 -
1 DATA STROBE		
2 DATA BIT 1		
3 DATA BIT 2		
4 DATA BIT 3		
5 DATA BIT 4		
6 DATA BIT 5		
7 DATA BIT 6		
8 DATA BIT 7		
9 DATA BIT 8		
10 ACKNOWLEDGE		
11 BUSY		
12 PAPER OUT or PRINT OFF		
13 PRINT ENABLE and NOT OUT OF PAPER		
14 GROUND		
15 -		
16 GROUND		
17 CHASSIS GROUND		
18 +5V		
19 -		
20 -		
21 -		
22 -		
23 -		
24 - TWISTED PAIR GROUND		
25 -		
26 -		
27 -		
28 -		
29 -		
30 -		
31 -		
32 -		
33 GROUND		
34 -		

Table 1. Centronics Parallel pin-assignments Only 10 signal lines are required for a basic interface.

HERE IS THE COMPLETE TABLE:

OUR COMPLETE HEX TABLE

0	00				
1	01				
2	02				
3	03				
4	04				
5	05				
6	06				
7	07				
8	08				
9	09				
10	0A				
11	0B				
12	0C				
13	0D				
14	0E				
15	0F				
16	10				
17	11				
18	12				
19	13				
20	14				
21	15				
22	16				
23	17				
24	18				
25	19				
26	1A				
27	1B				
28	1C				
29	1D				
30	1E				
31	1F				
32	20				
33	21				
34	22				
35	23				
36	24				
37	25				
38	26				
39	27				
40	28				
41	29				
42	2A				
43	2B				
44	2C				
45	2D				
46	2E				
47	2F				
48	30				
49	31				
50	32				
		51	33	101	65
		52	34	102	66
		53	35	103	67
		54	36	104	68
		55	37	105	69
		56	38	106	6A
		57	39	107	6B
		58	3A	108	6C
		59	3B	109	6D
		60	3C	110	6E
		61	3D	111	6F
		62	3E	112	70
		63	3F	113	71
		64	40	114	72
		65	41	115	73
		66	42	116	74
		67	43	117	75
		68	44	118	76
		69	45	119	77
		70	46	120	78
		71	47	121	79
		72	48	122	7A
		73	49	123	7B
		74	4A	124	7C
		75	4B	125	7D
		76	4C	126	7E
		77	4D	127	7F
		78	4E	128	80
		79	4F	129	81
		80	50	130	82
		81	51	131	83
		82	52	132	84
		83	53	133	85
		84	54	134	86
		85	55	135	87
		86	56	136	88
		87	57	137	89
		88	58	138	8A
		89	59	139	8B
		90	5A	140	8C
		91	5B	141	8D
		92	5C	142	8E
		93	5D	143	8F
		94	5E	144	90
		95	5F	145	91
		96	60	146	92
		97	61	147	93
		98	62	148	94
		99	63	149	95
		100	64	150	96
				151	97
				152	98
				153	99
				154	9A
				155	9B
				156	9C
				157	9D
				158	9E
				159	9F
				160	A0
				161	A1
				162	A2
				163	A3
				164	A4
				165	A5
				166	A6
				167	A7
				168	A8
				169	A9
				170	AA
				171	AB
				172	AC
				173	AD
				174	AE
				175	AF
				176	B0
				177	B1
				178	B2
				179	B3
				180	B4
				181	B5
				182	B6
				183	B7
				184	B8
				185	B9
				186	BA
				187	BB
				188	BC
				189	BD
				190	BE
				191	BF
				192	C0
				193	C1
				194	C2
				195	C3
				196	C4
				197	C5
				198	C6
				199	C7
				200	C8
				201	C9
				202	CA
				203	CB
				204	CC
				205	CD
				206	CE
				207	CF
				208	D0
				209	D1
				210	D2
				211	D3
				212	D4
				213	D5
				214	D6
				215	D7
				216	D8
				217	D9
				218	DA
				219	DB
				220	DC
				221	DD
				222	DE
				223	DF
				224	E0
				225	E1
				226	E2
				227	E3
				228	E4
				229	E5
				230	E6
				231	E7
				232	E8
				233	E9
				234	EA
				235	EB
				236	EC
				237	ED
				238	EE
				239	EF
				240	FO
				241	F1
				242	F2
				243	F3
				244	F4
				245	F5
				246	F6
				247	F7
				248	F8
				249	F9
				250	FA
				251	FB
				252	FC
				253	FD
				254	FE
				255	FF